Sri Lanka Institute of Information Technology

Bank Churn Intelligence
Customer Churn Prediction System

Project Report

Fundamentals of Data Mining Project (IT3051) - 2025

FDM_MLB_G16/ Mini Miners

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On
6th October 2025

Video link : Video Preview

Deployed link: View Deployment

Github link: GitHub Here

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1. Background

Customer churn is one of the most critical challenges in the banking industry, leading to significant financial loss and increased acquisition costs. Retaining existing customers is far more cost-effective than acquiring new ones, and therefore, predicting potential churners is essential for maintaining customer loyalty and improving business sustainability. This project aims to apply machine learning techniques to predict bank customer churn using demographic, financial, and behavioral attributes. The objective is to build an effective model that helps banking institutions identify high-risk customers, enabling proactive retention strategies and better decision-making.

2. Target and Business Goals

The main goals of this project are to:

- Accurately identify customers at high risk of churning.
- Provide data-driven insights into customer retention strategies.
- Enable the bank to take proactive actions to minimize churn.
- Support management decision-making through visual analytics and dashboards.
- Build a deployable churn prediction system that integrates with real-world data environments.

Fig: Illustration of customer churn in banking.

Fig: Machine learning-based churn prevention model.

3. Choice of Technology

Python was selected as the primary development language due to its rich ecosystem of data analysis and machine learning libraries such as Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn.

Flask was used to develop the RESTful API for generating predictions, while Streamlit was utilized for building the interactive dashboard.

We have deployed the solution on the AWS Free Tier, utilizing cloud infrastructure for scalability and reliable performance.

Libraries and frameworks used:

- Pandas, NumPy – Data manipulation and preprocessing
- Scikit-learn – Model training, evaluation, and prediction
- Matplotlib, Seaborn – Visualization
- Flask – Backend API
- Streamlit – Dashboard and reporting interface

4. Dataset Description

The dataset utilized for this project is a **large-scale customer churn dataset** obtained from Kaggle, containing **over 100,000 customer records** and **26 distinct attributes** that describe various aspects of each customer's demographics, financial behavior, and activity patterns. The data was collected from a **commercial bank in Botswana**, and each record corresponds to a unique customer profile.

This dataset was chosen due to its **comprehensive representation of customer characteristics**, allowing for accurate modeling of churn behavior. It serves as a robust foundation for applying data mining and machine learning techniques to predict which customers are at risk of leaving the bank.

The **target variable** in this dataset is **Exited**, which indicates whether the customer has left (1) or stayed (0) with the bank. The remaining 25 attributes serve as independent variables used to train and evaluate machine learning models.

4.1 Dataset Composition

The dataset contains a mixture of **numerical** and **categorical** variables, categorized into four main groups as follows:

A. Demographic and Identification Attributes

These features describe the customer's personal and regional details.

- **CustomerId** – Unique customer identifier (non-predictive).
- **Surname** – Customer's surname (excluded from modeling).
- **Gender** – Male or Female.
- **Age** – Numeric age of the customer.
- **Geography** – Customer's region (e.g., Botswana, Namibia, Zambia).
- **MaritalStatus** – Indicates whether the customer is Single, Married, or Divorced.
- **Dependents** – Number of dependents supported by the customer.

B. Financial and Account Attributes

These variables capture financial standing and engagement with bank products.

- **CreditScore** – Numeric indicator of creditworthiness.
- **Balance** – Average account balance.
- **EstimatedSalary** – Customer’s estimated annual income.
- **LoanAmount** – Total outstanding loan amount.
- **NumOfProducts** – Number of financial products owned by the customer.
- **HasCrCard** – Whether the customer holds a credit card (1 = Yes, 0 = No).
- **IsActiveMember** – Indicates customer activity level (1 = Active).

C. Transactional and Behavioral Attributes

These features track the customer’s activity and interaction with the bank.

- **Tenure** – Number of years with the bank.
- **TransactionCount** – Number of transactions made during the period.
- **TransactionFrequency** – Average monthly transaction frequency.
- **OnlineActivityScore** – Reflects digital banking activity (logins, online payments, etc.).
- **CustomerSatisfactionScore** – Overall customer feedback score on a scale from 1 to 5.
- **NumComplaints** – Number of recorded complaints.
- **SupportInteractions** – Frequency of customer support interactions.

D. Risk and Retention Indicators

These variables relate to customer risk levels and loyalty measures.

- **AccountType** – Type of account held (Savings, Current, Premium).
- **LoanDefault** – Indicates loan default history (1 = Yes, 0 = No).

- **CreditCardLimit** – The maximum spending limit on the credit card.
- **ActiveDays** – Days the customer has been active recently.
- **Exited (Target Variable)** – Indicates whether the customer churned (1) or stayed (0).

4.2 Summary of the Dataset

- **Number of Records:** 100,000+
- **Number of Attributes:** 26
- **Feature Types:** 17 Numerical, 9 Categorical
- **Target Variable:** Exited (binary classification: churned or retained)
- **Missing Values:** Minimal, handled using median or mode imputation.
- **Class Distribution:** Approximately 23% churned vs. 77% retained customers.

The dataset offers an extensive range of information required for analyzing and predicting churn behavior. Its diverse mix of demographic, transactional, and engagement variables enables the development of a comprehensive and explainable churn prediction model

5. Data Preprocessing and Transformation

This section explains the steps followed to clean, prepare, and analyze the dataset before training machine learning models. Proper preprocessing ensures the dataset is consistent and ready for reliable model performance.

5.1 Data Cleaning

```
[5]
✓ 0s
print(df.isnull().sum())

RowNumber                  137
customer_id                135
Surname                  13495
first_name                6301
dob                        142
Gender                     138
marital_status             136
dependents                 439
Occupation                9228
Income                    7248
education                  141
Address                   8626
phone                    14598
tenure_years               144
segment                    137
preferred_contact          134
credit_score              8750
credit_history_years       139
outstanding_debt          3094
churned                    140
churn_reason            103189
churn_date              103184
Balance                   5454
products_count             137
complaints_count           140
dtype: int64
```

Fig : Missing value checks

```
df.columns = df.columns.str.strip().str.lower().str.replace(' ', '_')   #clean column names

df.drop(['churn_reason', 'churn_date'], axis=1, inplace=True)   # Drop Irrelevant / High-Null Columns

df.dropna(subset=['customer_id', 'rownumber'], inplace=True)   # Handle Missing Values
```

Fig : Includes basic data preparation and structural fixes

5.2 Data Type Conversion and Handling

```
[9]
# Fill categorical

cat_cols = ['gender', 'marital_status', 'dependents', 'occupation', 'education',
            'address', 'phone', 'segment', 'preferred_contact', 'credit_history_years',
            'churned', 'products_count', 'complaints_count']
for col in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])
```

```
[10]
# Fill numerical
num_cols = ['credit_score', 'outstanding_debt', 'income', 'balance', 'tenure_years']
for col in num_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')  # convert to numeric
    df[col] = df[col].fillna(df[col].median())
```

```
[11]
df.drop_duplicates(inplace=True)    # Remove duplicates
```

```
[13]
# Process Date Column

from datetime import datetime

df['dob'] = pd.to_datetime(df['dob'], errors='coerce')
df['age'] = df['dob'].apply(lambda x: datetime.now().year - x.year if pd.notnull(x) else None)
```

```
[14]
# drop dob

df.drop('dob', axis=1, inplace=True)   # bcz of becomes redundant
```

Fig : Ensures all data is in the correct format for analysis.

5.3 Outlier Detection and Treatment

```
[12]
# Handle Outliers in Numerical Columns

def cap_outliers(df, column):
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    df[column] = df[column].clip(lower_bound, upper_bound)
    return df

# Apply to relevant numerical columns
num_cols = ['credit_score', 'outstanding_debt', 'income', 'balance', 'tenure_years']
for col in num_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')  # ensure numeric
    df = cap_outliers(df, col)
```

Fig : Improves model performance by reducing the impact of extreme values.

5.4 Feature Transformation

```
[15]
✓ 0s
# Feature Scaling

from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
scaled_cols = ['credit_score', 'outstanding_debt', 'income', 'balance', 'tenure_years', 'age']
df[scaled_cols] = scaler.fit_transform(df[scaled_cols])
```

```
[16]
✓ 0s
# Encode Categorical Features

from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()
df['gender'] = le.fit_transform(df['gender'])
df['churned'] = le.fit_transform(df['churned'].astype(str))
```

Fig : Prepares the data for modeling through scaling and encoding.

5.5 Final Data Preparation

```
[17]
✓ 2s
# save preprocessed csv

import os

os.makedirs("data/processed", exist_ok=True)
df.to_csv("Customer_churn/data/churn_cleaned.csv", index=False)
```

Fig : Creates the cleaned and ready-to-use dataset for modeling.

5.6 Exploratory Data Analysis (EDA)

The Exploratory Data Analysis (EDA) phase was carried out to understand the structure, distribution, and relationships within the cleaned dataset.
It helps identify the factors that influence customer churn and guides the selection of features for model building.

5.6.1 Target Variable Distribution (Churn Rate)

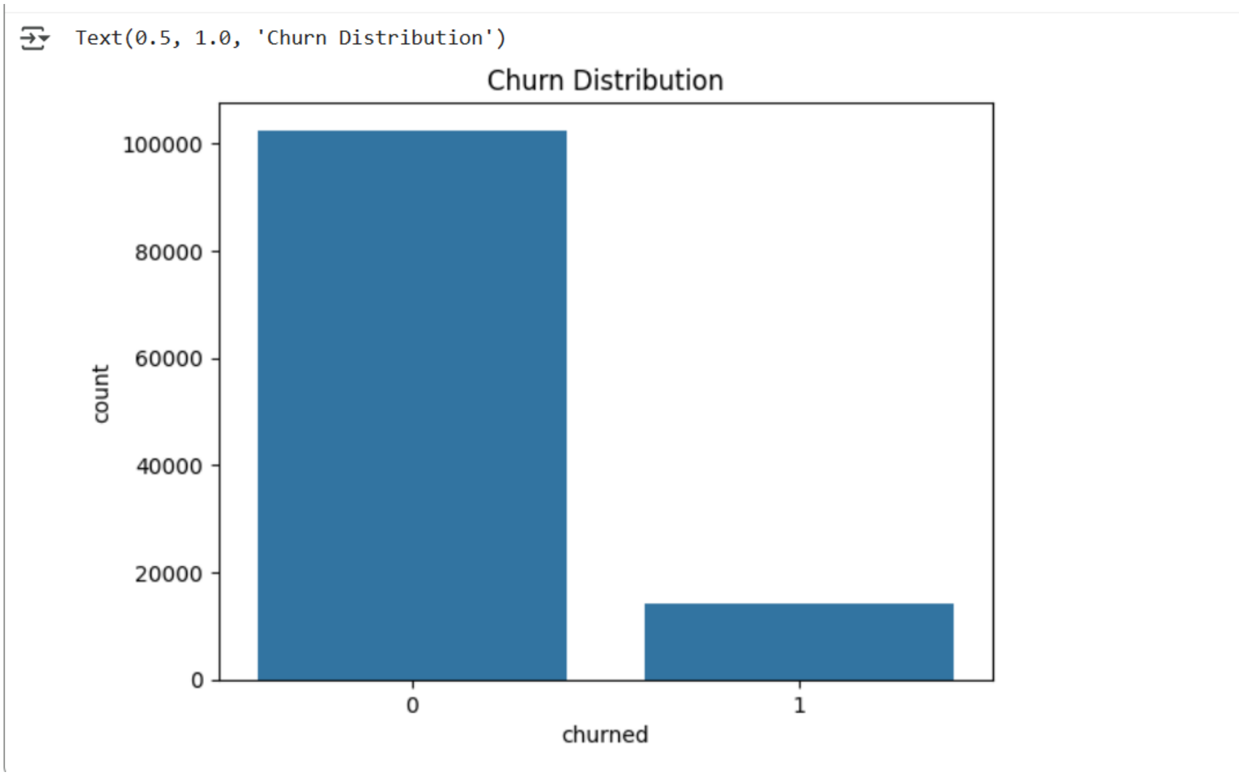


Fig : The dataset showed more non-churn customers than churned ones, indicating an imbalance important for model training.

5.6.2 Numerical Feature Analysis

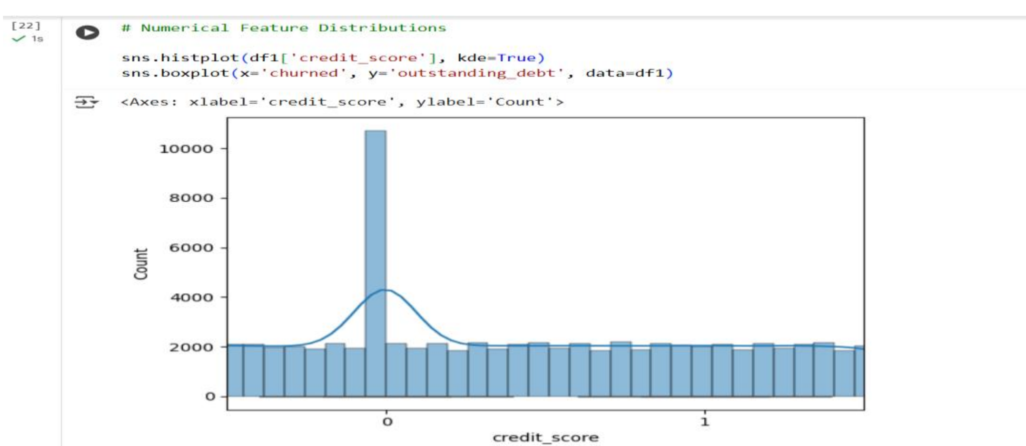


Fig : Credit_score vs Count

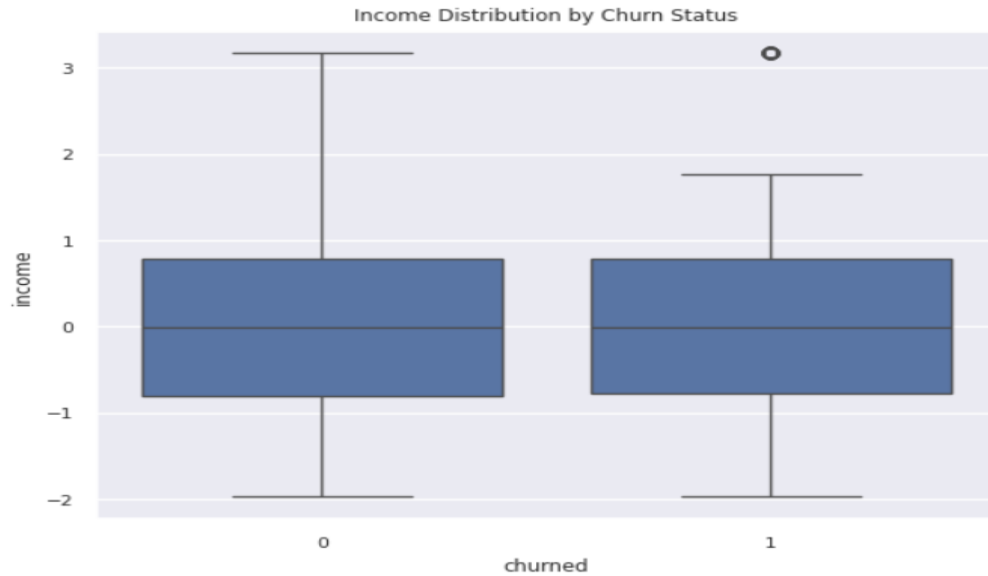


Fig : Income Distribution vs Churn Status

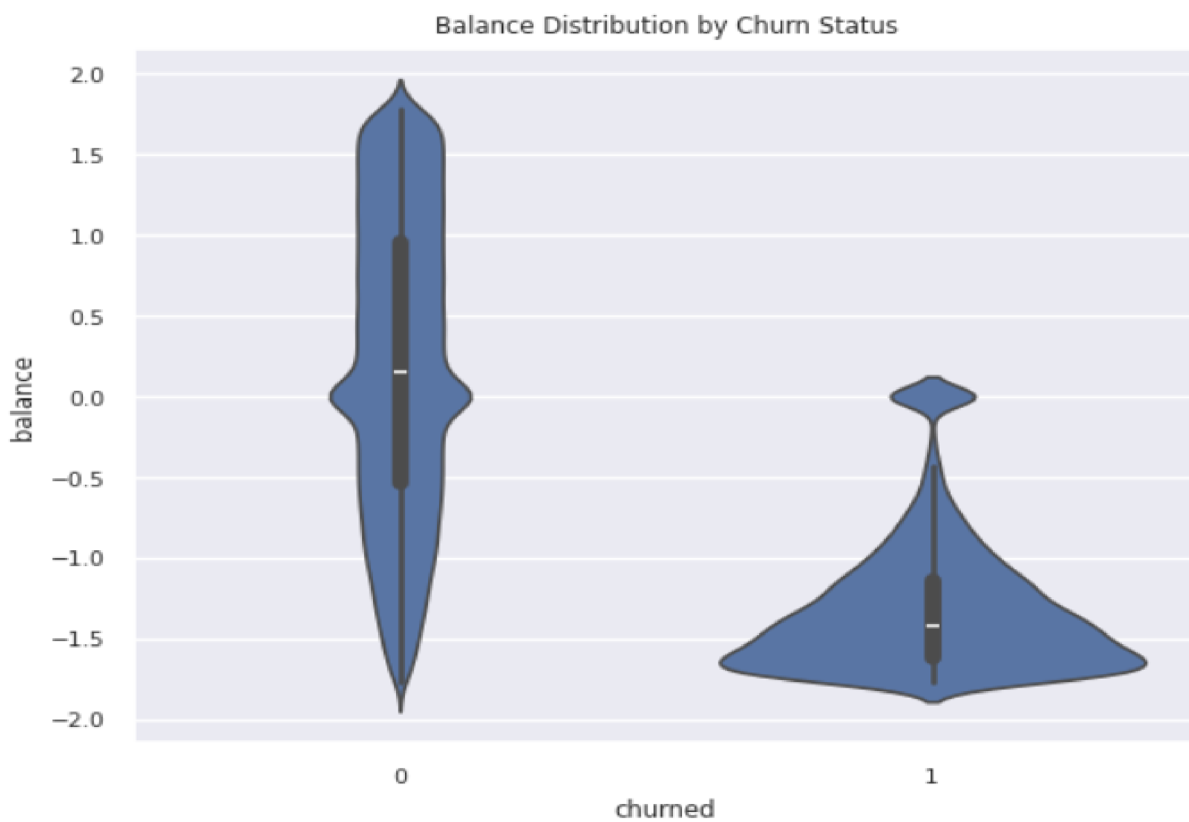


Fig : Balance Distribution vs Churn Status

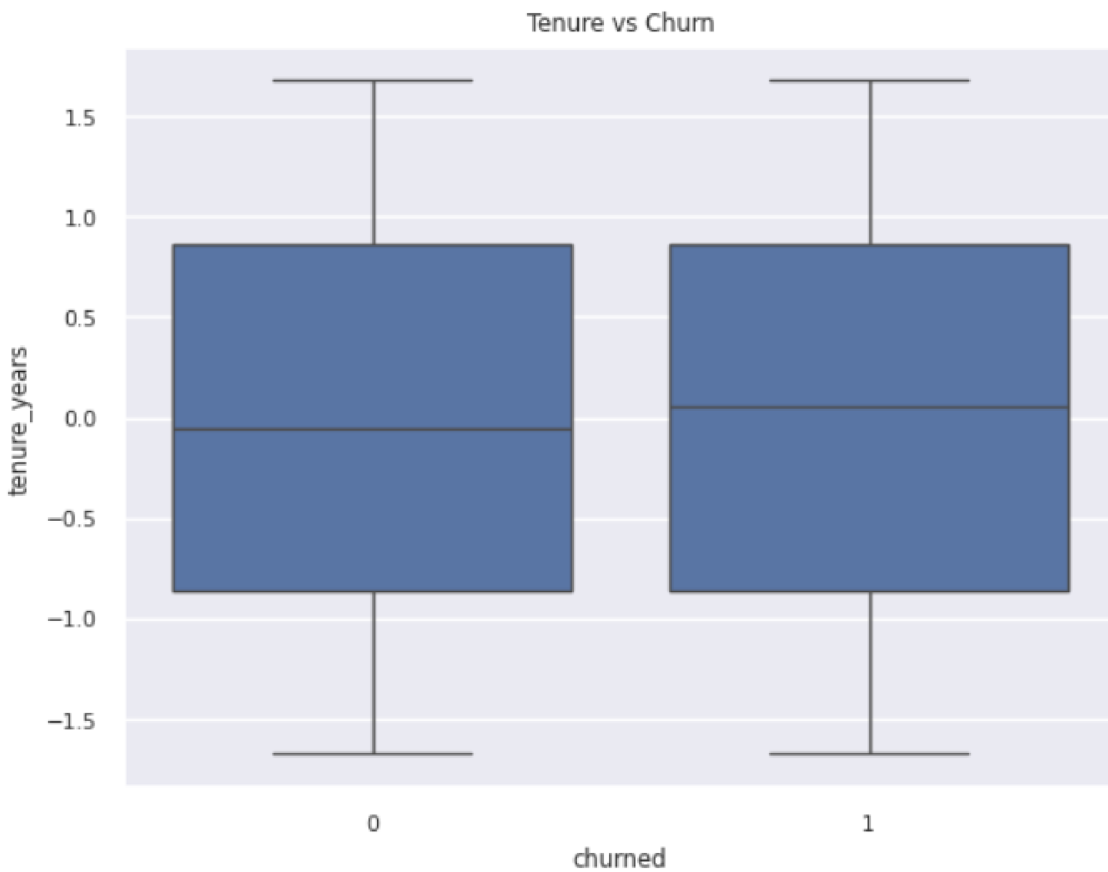


Fig : Tenure vs Churn Status

5.6.3 Categorical Feature Analysis

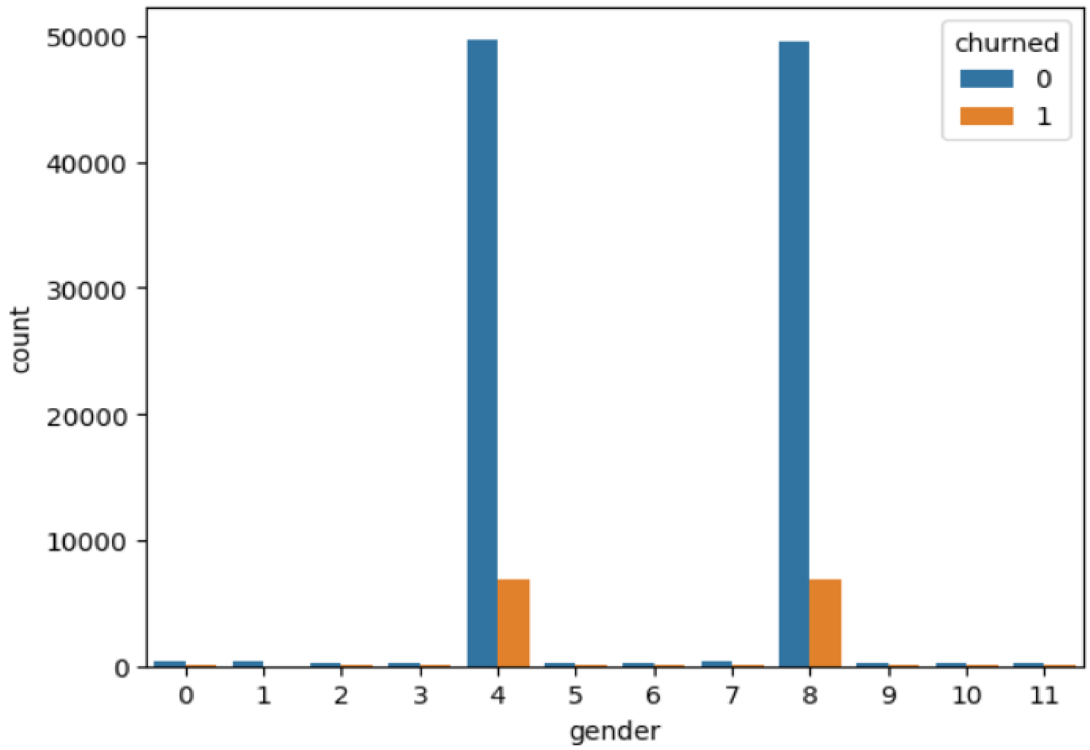


Fig : Gender vs Count

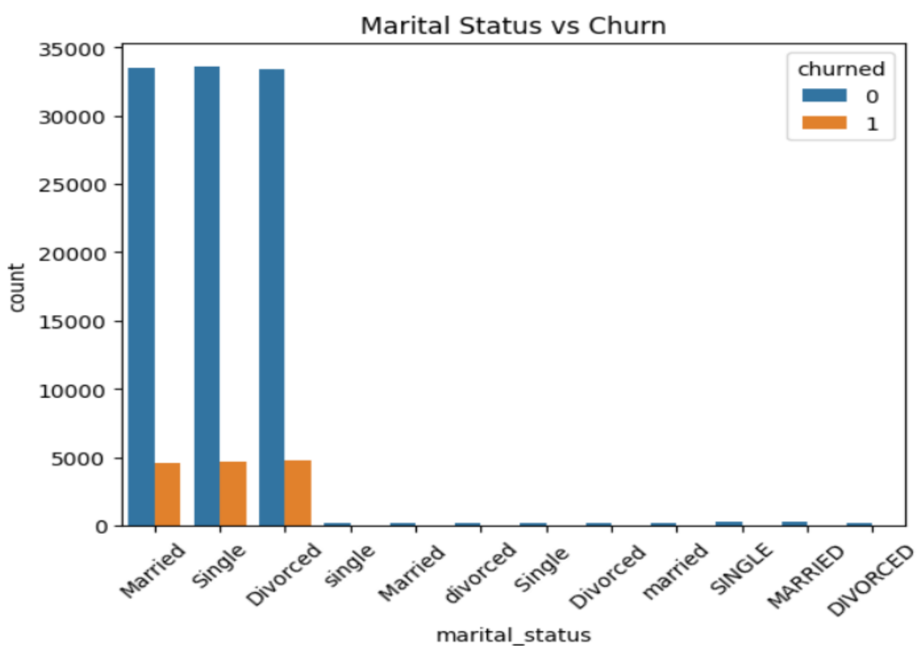


Fig : Marital Status vs Churn

5.6.4 Correlation Analysis

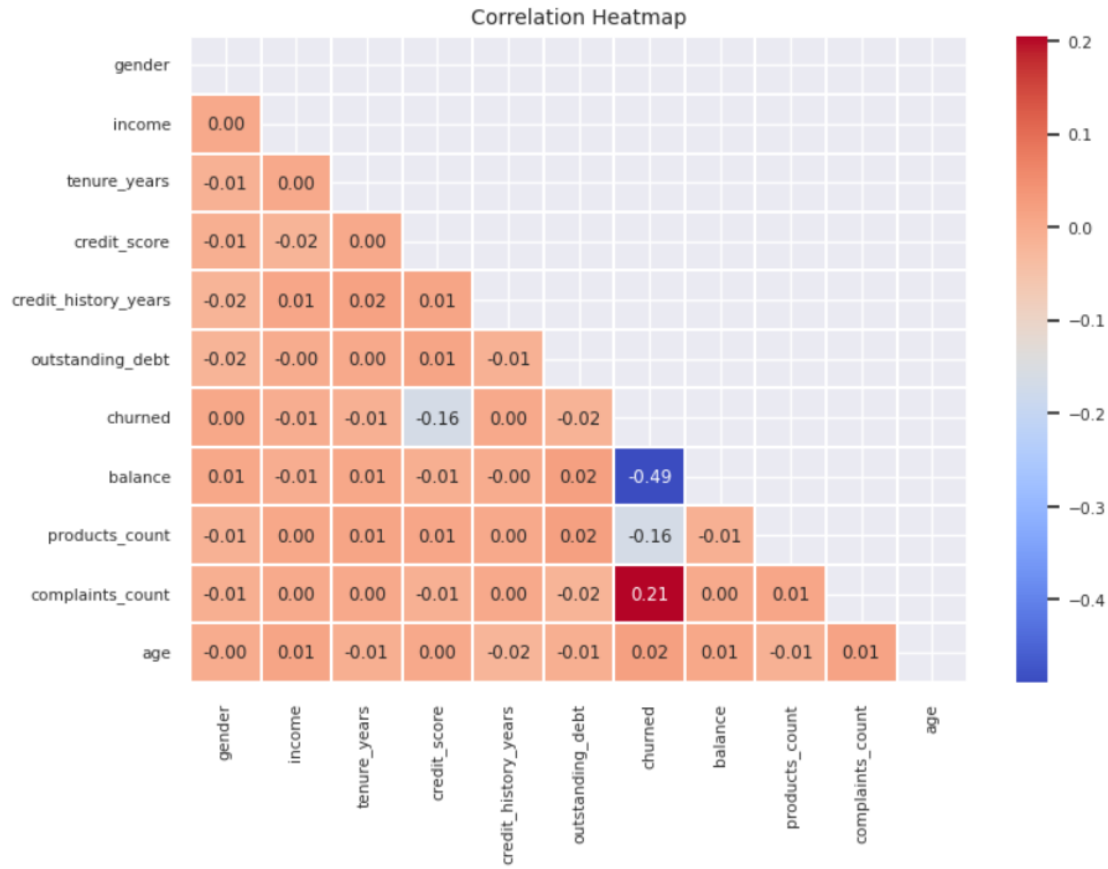


Fig : Correlation Heatmap

5.6.5 Feature Importance Analysis

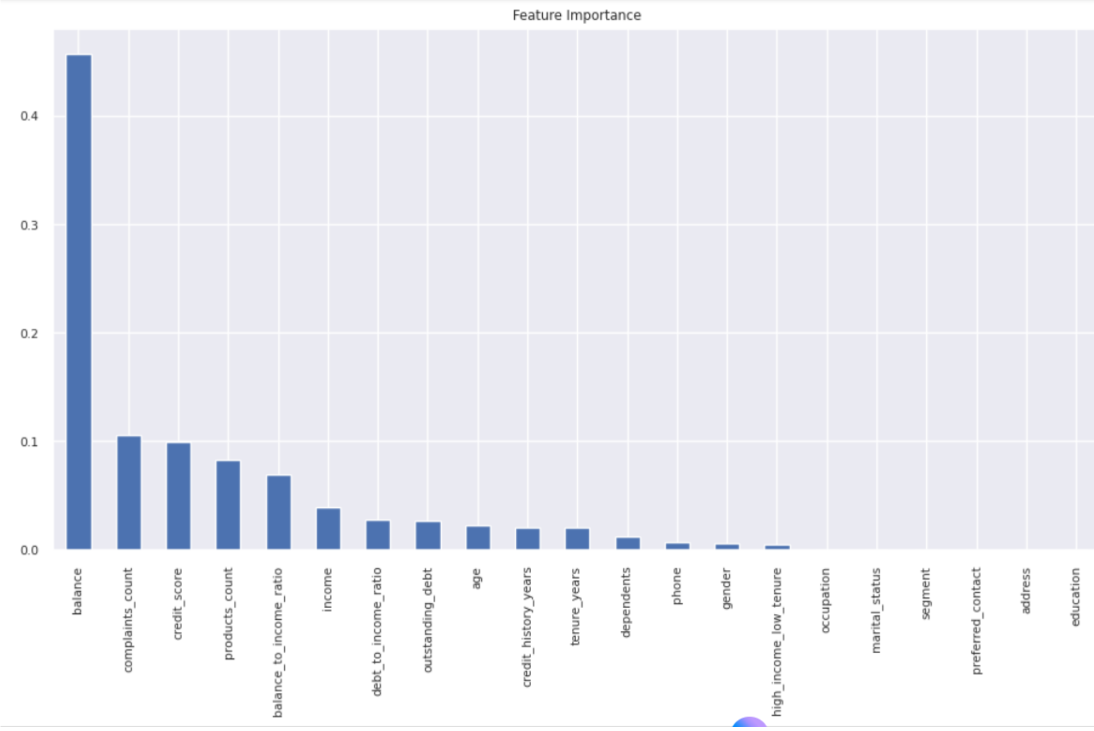


Fig : Feature Importance Analysis

5.6.6 Summary of EDA Findings

The EDA revealed that churned customers generally have **lower income**, **shorter tenure**, and **more variation in account balance**.
There are visible churn differences across **marital status, gender**, and possibly **customer segments and occupations**.
The correlation heatmap showed a **moderate negative relationship between balance and churn**, while other features showed weaker correlations.
Overall, these insights highlight key behavioral and financial factors influencing churn and help guide feature selection for model development.

6. Model Implementations

The project implemented several supervised machine learning algorithms to identify the best-performing model. The models considered included **Random Forest** and **XGBoost**. Each model was trained using an **80/20 train-test split** and evaluated with **Stratified 5-Fold Cross-Validation** to ensure reliable and unbiased performance. Hyperparameter tuning was applied to optimize model accuracy and generalization. Model performance was assessed using **Accuracy**, **Precision**, **Recall**, and **F1-score** metrics. Among the models, **XGBoost achieved the highest overall performance**, demonstrating strong predictive capability and effective handling of imbalanced data.

6.1 Random Forest Model Results

```
Output   Debug Console   Terminal   Ports
(base) PS C:\Users\Admin\Desktop\churnbank> powershell -NoProfile -
ExecutionPolicy Bypass -Command "cd 'C:\Users\Admin\Desktop\churnba
nk'; python rf_evaluate.py"
=== Random Forest Evaluation ===
Samples: train=82251, test=20563
Accuracy:      0.9799
Precision:     0.9379
Recall:        0.8966
F1-score:      0.9168
ROC-AUC:       0.9945
Confusion Matrix (TN, FP / FN, TP):
[[17868   151]
 [  263  2281]]
              precision    recall  f1-score   support

           0     0.9855    0.9916    0.9885     18019
           1     0.9379    0.8966    0.9168      2544

    accuracy                         0.9799     20563
   macro avg     0.9617    0.9441    0.9527     20563
weighted avg     0.9796    0.9799    0.9797     20563

(base) PS C:\Users\Admin\Desktop\churnbank>
```

fig : Random Forest A Results

```
>>
Accuracy: 0.9739141608840914

Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.99      0.99     20561
           1       0.95      0.82      0.88      2785

    accuracy                           0.97     23346
   macro avg       0.96      0.91      0.93     23346
weighted avg       0.97      0.97      0.97     23346


Model saved as 'random_forest_model.pkl'
(churn) PS C:\Users\manee\Customer_churn\model>
```

fig: Random Forest B Results

6.2 XGBoost Model Results

```
📊 Classification Report:
              precision    recall  f1-score   support

           0       0.99      0.99      0.99     20561
           1       0.91      0.91      0.91      2785

    accuracy                           0.98     23346
   macro avg       0.95      0.95      0.95     23346
weighted avg       0.98      0.98      0.98     23346


🧮 Confusion Matrix:
 [[20306   255]
 [  258  2527]]

🏆 Best Parameters: {'learning_rate': 0.1, 'max_depth': 4, 'n_estimators': 100}

✅ Model saved to model/xgb_churn_model.pkl

📈 Train Accuracy: 0.9803237067597588
📉 Test Accuracy: 0.9780262143407864
📊 Cross-validated Accuracy: 0.9820181795139806
```

fig: XGBoost Model Results

6.3 Model Performance Comparison

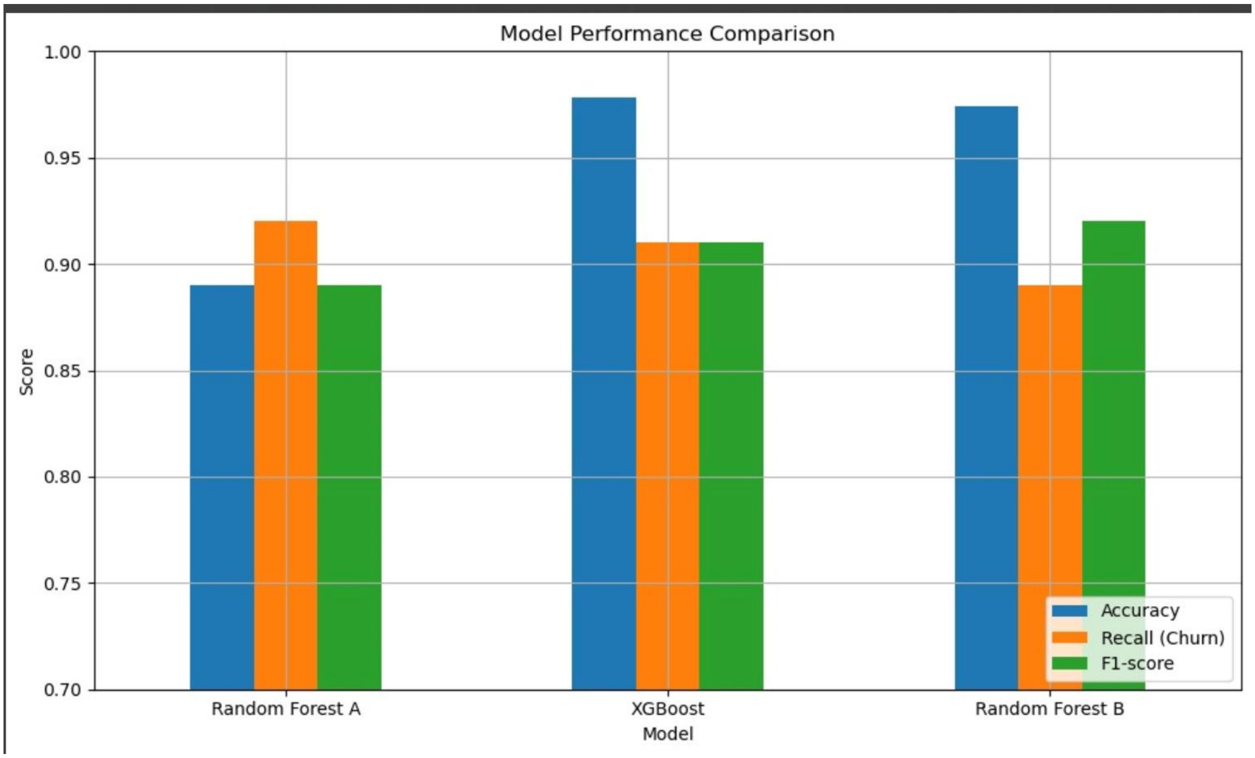


Fig : Model Performance Comparison

This bar graph compares the performance of the three tested models — **Random Forest A**, **XGBoost**, and **Random Forest B** — across key metrics such as **Accuracy**, **Recall**, and **F1-score**.
While both Random Forest models performed well, **XGBoost achieved the highest accuracy and the most balanced precision–recall performance.**

Based on these results, **XGBoost was selected as the final model** for deployment, as it provided the most reliable and consistent predictions for customer churn.

7. Results and Evaluation

This section presents the **results of the deployed churn prediction application**.
The app was designed to take user input, process it through the trained XGBoost model, and display whether a customer is likely to churn or not.
The following screenshots show the app's interface and its prediction outputs for different input cases.

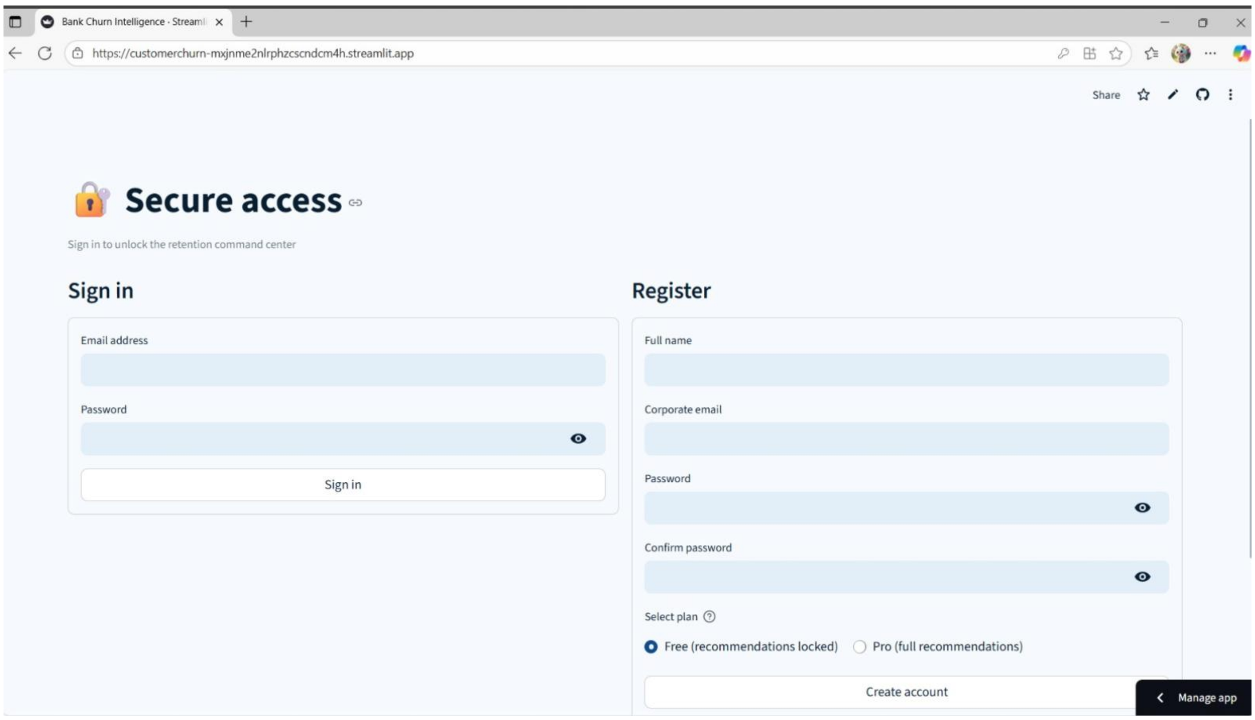


Fig : Secure user Login Interface to the app

7.1 Single User Prediction Scenario

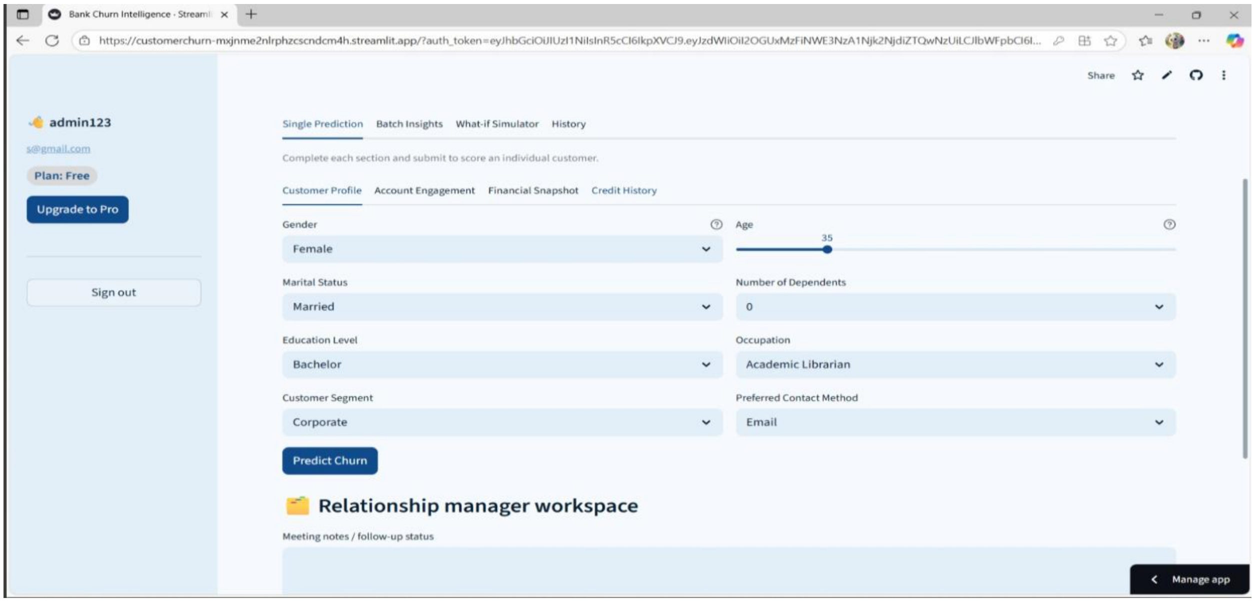


Fig : Application Interface

This screenshot shows the main interface of the **Customer Churn Prediction App**. Users can enter customer details such as credit score, age, balance, tenure, and number of products.

Once all fields are filled, the user can click the **"Predict Churn"** button to generate the churn prediction.

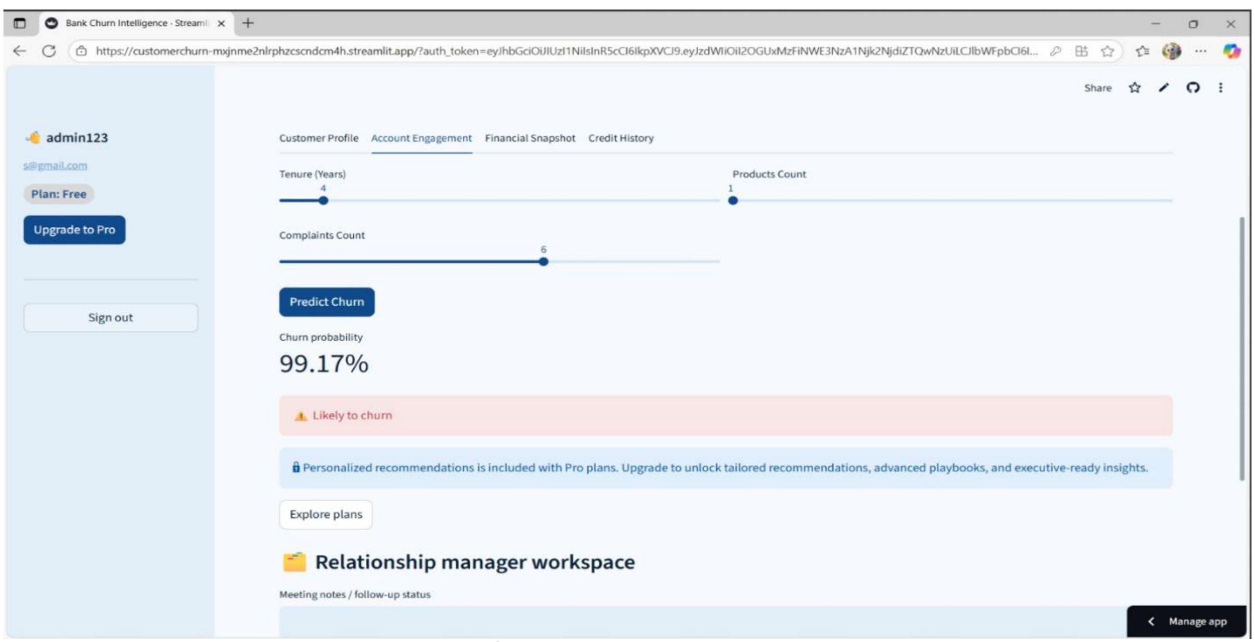


Fig : Output to the above input

7.2 Batch Prediction Functionality

In addition to single-customer prediction, the system also provides a **batch prediction feature**.
This allows users to upload a **CSV file** containing details of multiple customers at once. The system processes the entire file through the trained **XGBoost model** and generates churn predictions for each customer in the dataset.

Users can download a **CSV output file** that includes the original customer details along with the model's churn prediction results.
This feature is especially useful for banks or organizations that need to analyze churn risk for a large group of customers efficiently.

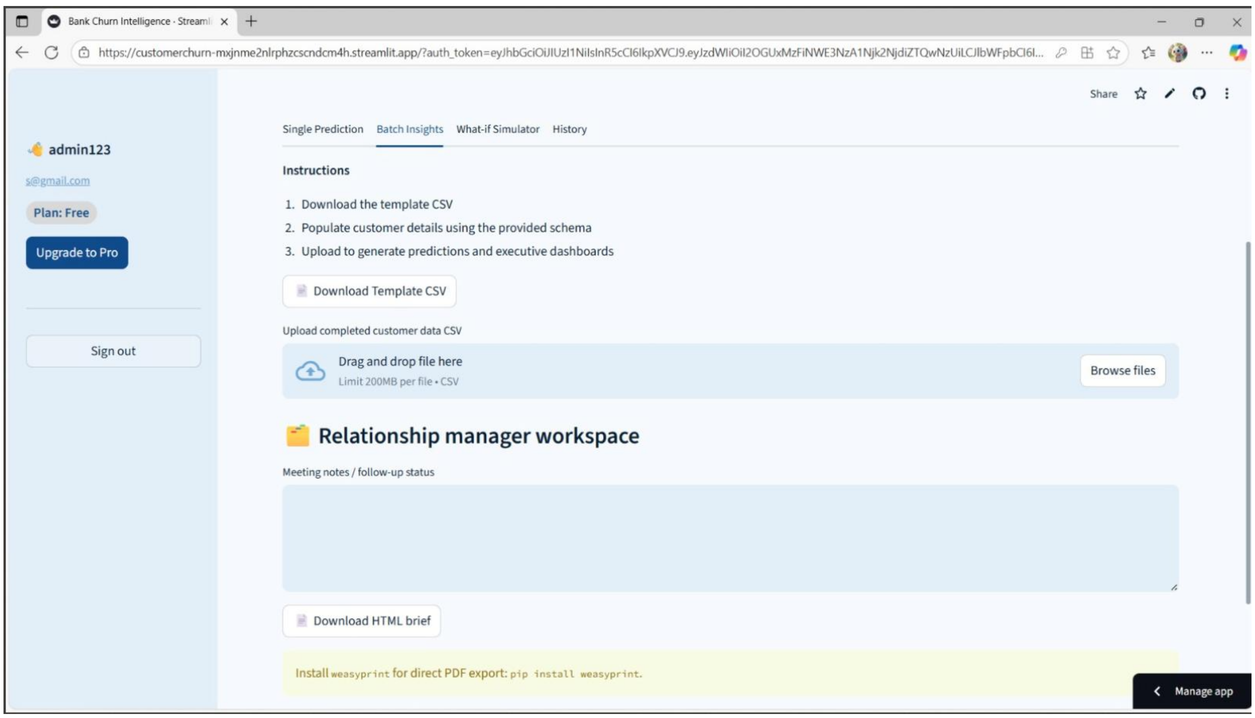


Fig : Batch prediction interface

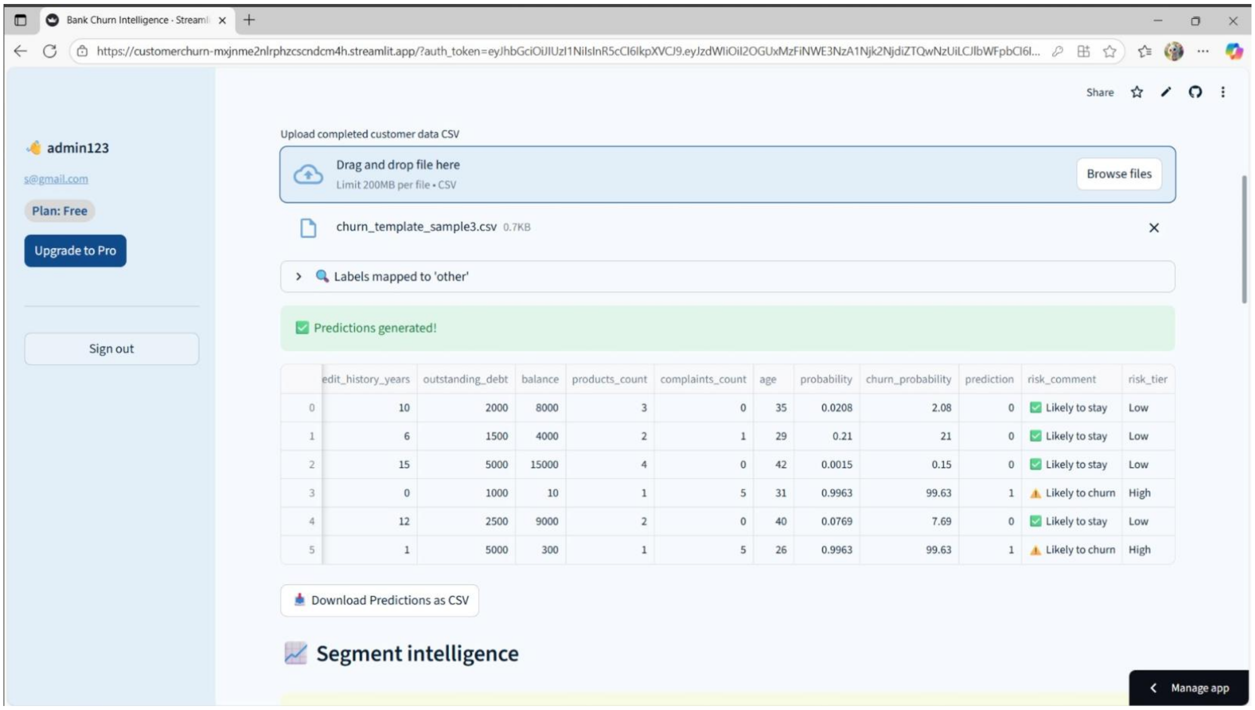


Fig : output generated after performing batch prediction.

8. Visualization and Insights

This section focuses on the **visual outputs and actionable insights** generated by the Customer Churn Prediction System.
When a user enters customer information or uploads a batch file, the system not only predicts churn but also generates **interactive visualizations** to help interpret the results.
Additionally, a **Pro User Functionality** provides personalized recommendations on actions to reduce churn risk, based on the model's analysis.

8.1 Visualizations for User Scenarios

The system dynamically generates visual graphs based on the input data provided by the user.
These visualizations help users quickly understand key factors influencing churn.

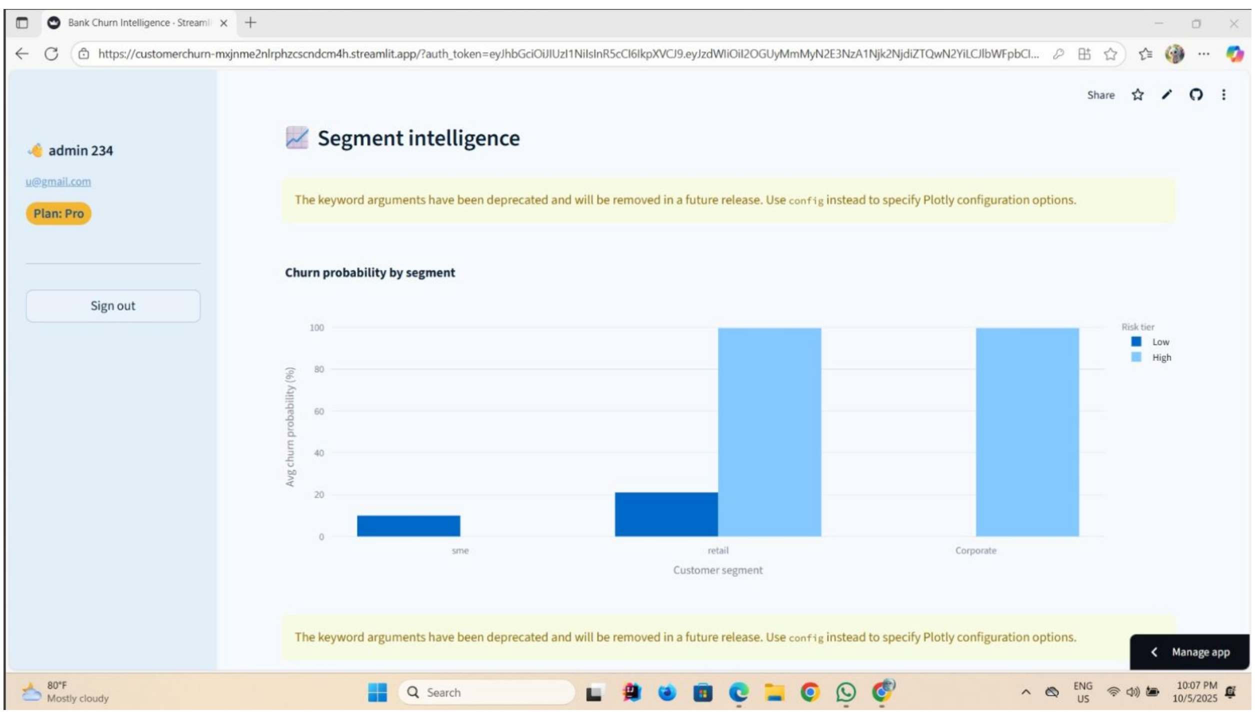


Fig : Churn Probability by Segment Chart

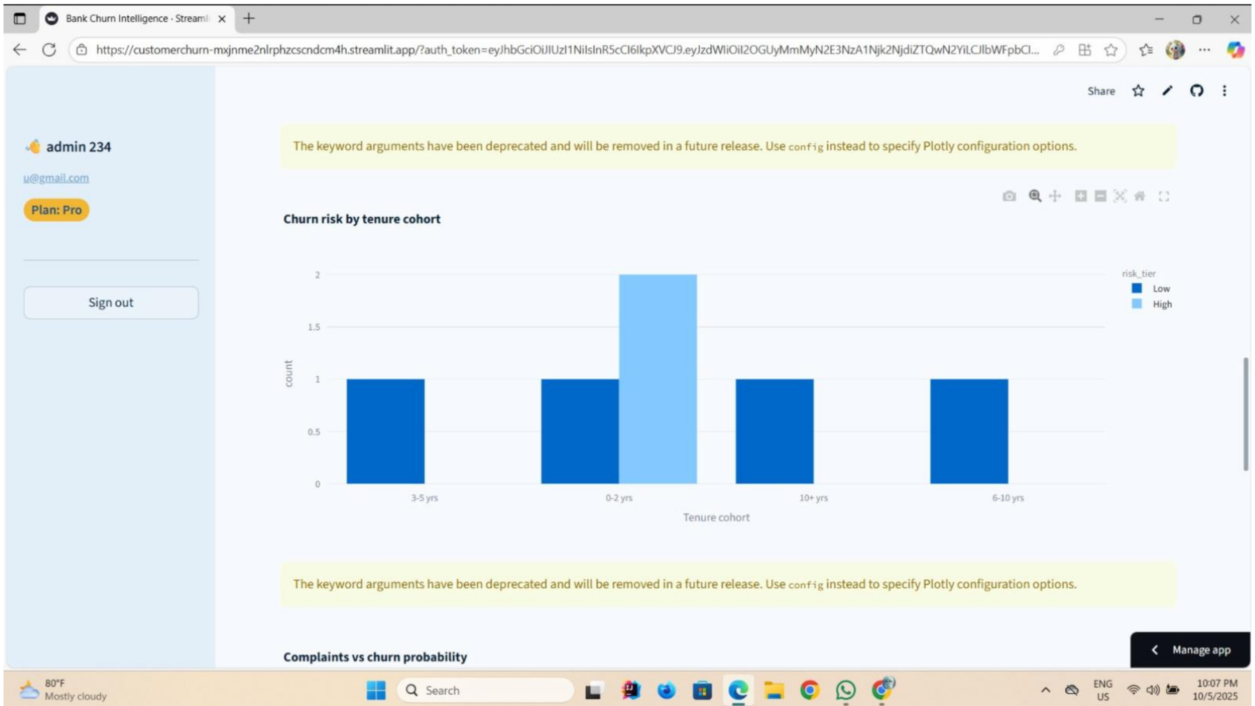


Fig : Churn Risk by Tenure Cohort Chart

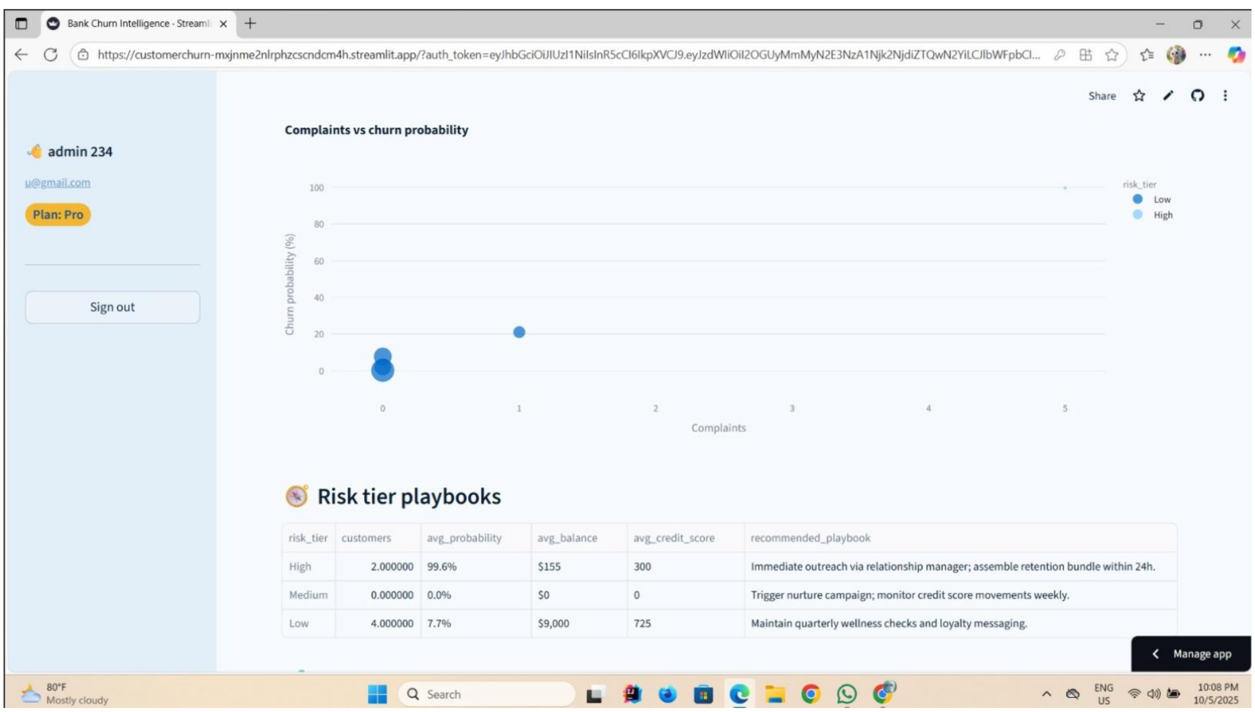


Fig : Complaints vs Churn Probability Chart

8.2 Pro User Recommendations

For users with **Pro Access**, the system offers **personalized recommendations** based on the prediction results.
If a customer is predicted to churn, the system analyzes the key risk factors and suggests targeted actions to help retain them.

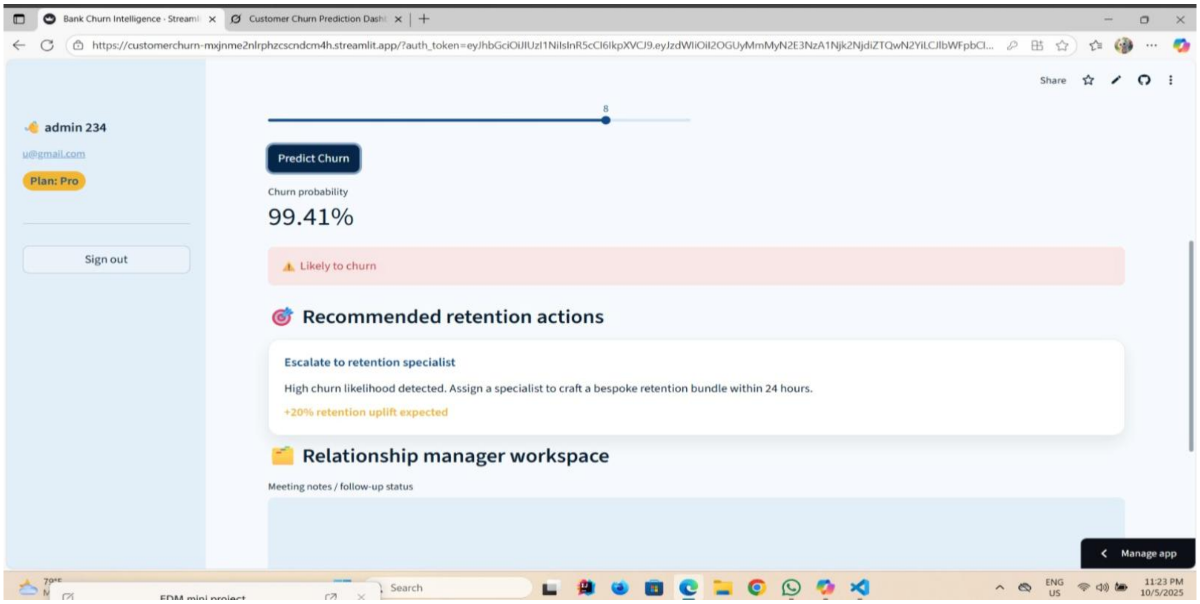


Fig : Recommendations for a single user

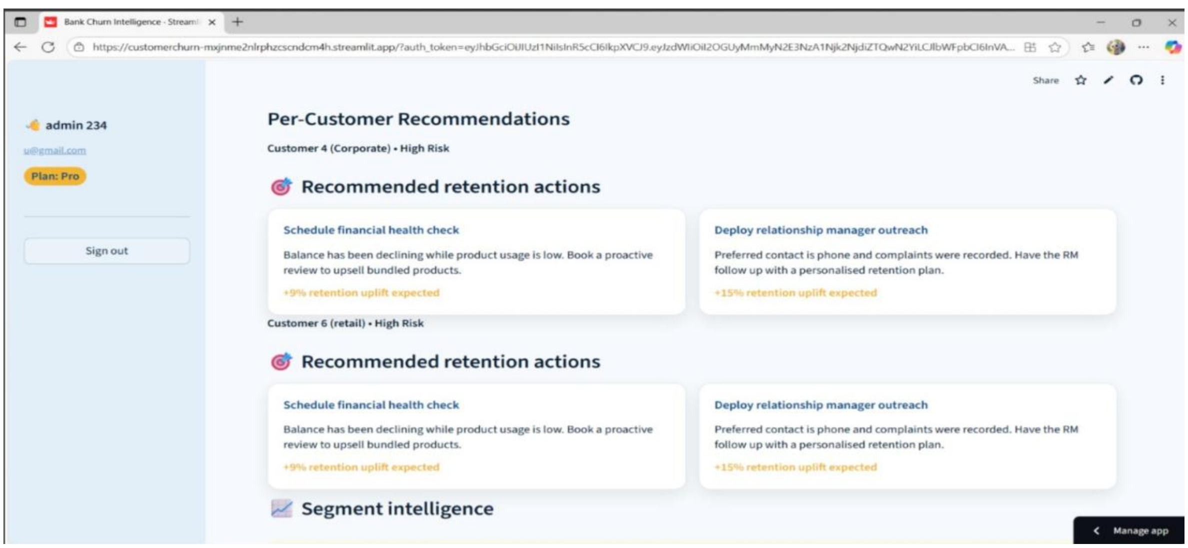


Fig : Batch User Recommendations for a Given Scenario

8.3 User History and Activity Tracking

The system includes a **User History** feature that allows users to view their previous predictions and actions.
Each time a prediction is made, the **input details, results, and recommendations** are saved for future reference.
Users can easily revisit past single or batch predictions to track trends and review churn outcomes.

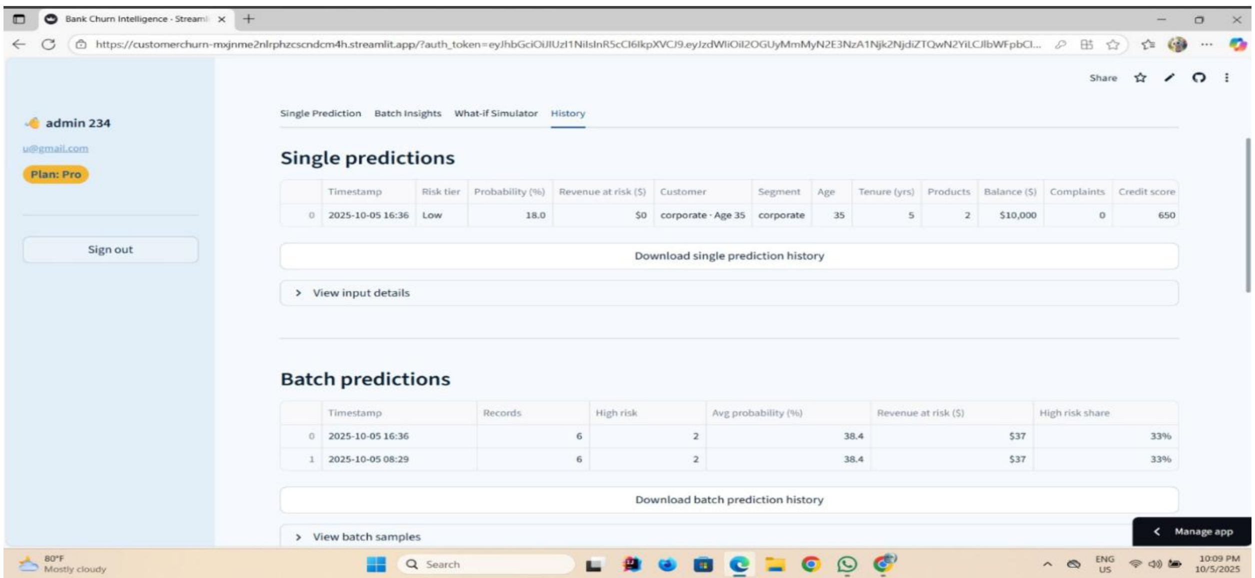


Fig : User History Page

8.4 What-If Simulator

The **What-If Simulator** allows users to explore how changes in customer attributes affect churn risk — without modifying the actual data.
Users can adjust variables such as **credit score, balance, or tenure** and instantly see how the predicted churn probability changes.
This helps relationship managers test different scenarios and understand which factors have the greatest impact on customer retention.

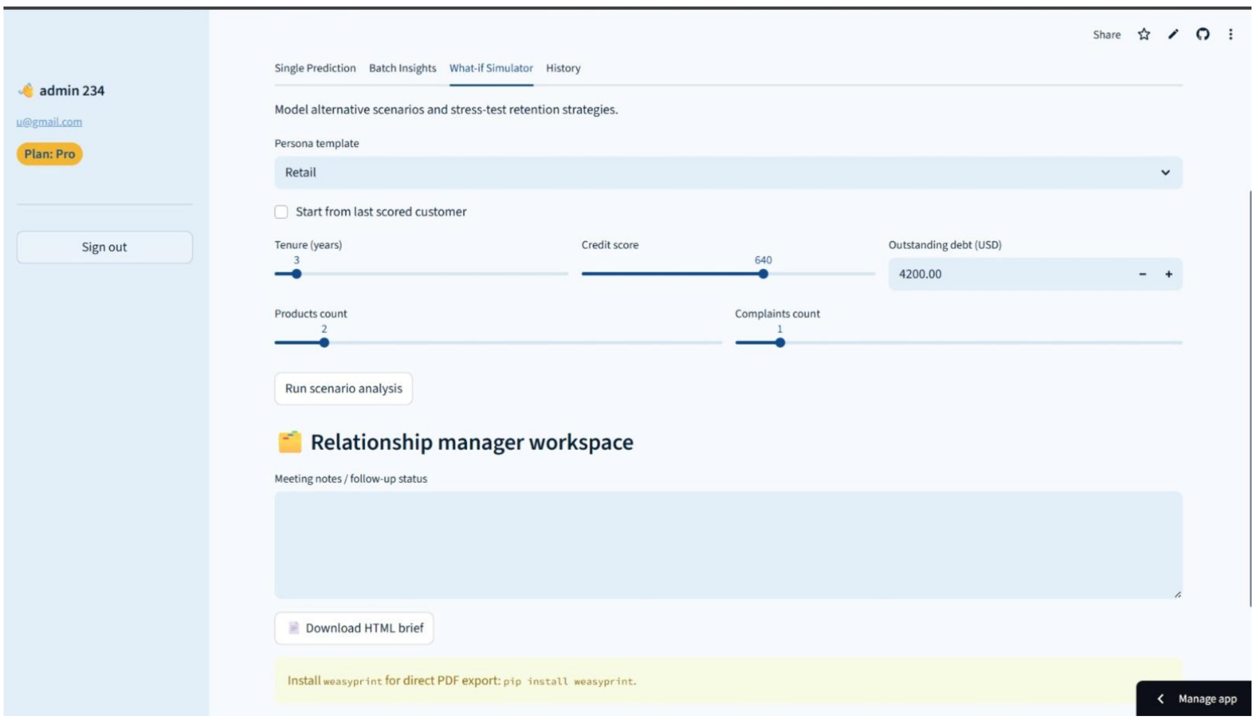


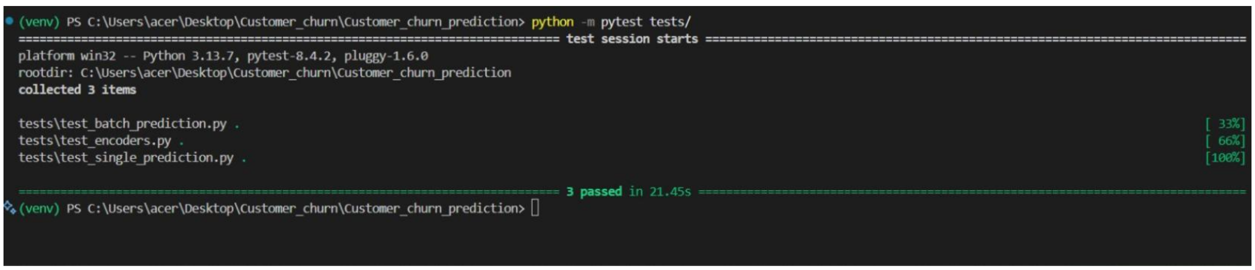
Fig : What-If Simulator

9. Testing

To ensure the reliability and correctness of the machine learning pipeline, we implemented a series of automated tests using **Pytest**. The testing process covered key components of the system, including data preprocessing, encoding functions, and both single and batch prediction modules.

All tests were executed successfully in the virtual environment using the command:

python -m pytest tests/



The test suite included three main test scripts — test_batch_prediction.py, test_encoders.py, and test_single_prediction.py. Each test validated the expected behavior and accuracy of the corresponding component.

The results showed that all tests passed successfully, confirming that the model's prediction functions, data transformations, and batch processing pipelines are working as intended. This ensures the overall stability and reliability of the churn prediction system.

10. Deliverables

The following deliverables were produced:

- Final project report documenting all methodologies and results.
- Trained machine learning models and evaluation metrics.
- RESTful API for generating churn predictions.
- Interactive Streamlit dashboard for visualization.
- Video presentation demonstrating model functionality and insights.

11. References

[1] S. D. Mfazi, *Bank Customer Churn Dataset*, Kaggle, 2023. [Online]. Available: https://www.kaggle.com/datasets/sandiledesmondmfazi/bankcustomer-churn [Accessed: Oct. 5, 2025].

[2] Data Science for All, *Customer Churn Prediction Project Tutorial*, YouTube, 2023. [Online Video]. Available: https://youtu.be/Ixe1kcYTSyo?si=wAYLAHUO6L0DdEH5 [Accessed: Oct. 5, 2025].